From these two figures it is clear that the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = x^{2n+1}, n \in \mathbb{N}$ is one-to-one and onto and hence its inverse function from $\mathbb{R}$ onto $\mathbb{R}$ exists. But $f : \mathbb{R} \to [0, \infty)$ given by $f(x) = x^{2n}, n \in \mathbb{N}$ is onto but not one-to-one. However, f is one-to-one and onto if we restrict its domain to $[0, \infty)$. This is helpful in understanding nth root of a real number. So we have two cases;

Case 1 When n is even.

In this case $y^n = a$ is not meaningful when $a < 0$. So no such y exists when $a < 0$.

Assume that $a > 0$. If y is a solution to $x^n = a$, then $-y$ is also solution to $x^n = a$.

Case 2 When n is odd.

In this case no such problem arises as in Case 1. For $y \in \mathbb{R}$, there is a unique $x \in \mathbb{R}$ such that $y = x^n$.

Based on the above observation we define radicals as follow.

Definition 2.2

(i) For $n \in \mathbb{N}$, n even, and $b > 0$, there is a unique $a > 0$ such that $a^n = b$.

(ii) For $n \in \mathbb{N}$, n odd, $b \in \mathbb{R}$, there is a unique $a \in \mathbb{R}$ such that $a^n = b$. In both cases a is called the nth root of b or radical and is denoted by $b^{1/n}$ or $\sqrt[n]{b}$



(i) If $n = 2$, then nth root is called the square root; if $n = 3$, then it is called cube root.

(ii) Observe that the equation $x^2 = a^2$, has two solutions $x = a$, $x = -a$; but $\sqrt{a^2} = |a|$.

(iii) Properties of exponents given above are still valid for radicals provided each of the individual terms are defined.

(iv) Note that for $n \in \mathbb{N}$ and $a \neq 0$ we have

$$(a^n)^{1/n} = \begin{cases} |a| & \text{if } n \text{ is even}, \\ a & \text{if } n \text{ is odd}. \end{cases}$$

For example, $\sqrt[4]{(-2)^4} = 16^{1/4} = 2$, $343^{1/3} = 7$ and $(-1000)^{\frac{1}{3}} = -10$.

For any rational $r = \dfrac{m}{n}$, $m \in \mathbb{Z}$, $n \in \mathbb{N}$, with $\gcd(m, n) = 1$ and for $a > 0$ we define $a^r = a^{\frac{m}{n}} = (a^{1/n})^m$.

For example, $49^{3/2} = (49^{1/2})^3 = 7^3 = 343$. But $(-49)^{3/2}$ has no meaning in real number system because there is no real number x such that $x^2 = (-49)^3$.

2.8.3 Exponential Function

Observe that for any $a > 0$ and $x \in \mathbb{R}$, a^x can be defined. If $a = 1$, we define $1^x = 1$. So we shall consider a^x, $x \in \mathbb{R}$ for $0 < a \neq 1$. Here a^x is called **exponential function with base** a. Note that a^x may not be defined if $a < 0$ and $x = \dfrac{1}{m}$ for even $m \in \mathbb{N}$. This is why we restrict to $a > 0$. Also, $a^x > 0$ for all $x \in \mathbb{R}$. It does also satisfy the following:

Properties of Exponential Function

For $a, b > 0$ and $a \neq 1 \neq b$

(i) $a^{x+y} = a^x\, a^y$ for all $x, y \in \mathbb{R}$,

(ii) $\dfrac{a^x}{a^y} = a^{x-y}$ for all $x, y \in \mathbb{R}$,